

Camunda 8 — Test Strategy & Results

Project: ProBuild Supplies Ltd Process Automation **Module:** UFCFAF-30-3 Development of Information Systems Project **Engine:** Camunda 8.9.0 self-managed (local) **Test date:** 2026-05-06 / 2026-05-07 **Tester:** Daniel Vasilev (Engineer 5 — Integration & Orchestration Lead)

1. Test Environment

Component	Detail
Process engine	Camunda 8.9.0 self-managed (local Docker)
Zeebe broker	localhost:26501 (gRPC)
REST API	localhost:8080
Tasklist	localhost:8082
Operate	localhost:8081
Worker runtime	Java 21 + Spring Boot 3, Maven
Worker entry point	WorkerApplication.java (mvn spring-boot:run)
Registered workers	51 @JobWorker methods across 5 @Component classes
BPMN under test	operational BPMN V2.bpmn (version 6, deployed 2026-05-06)
Executable processes	Pro (ProBuild), Process_0c6d9wt (FixPro), Process_1hz3vp5 (customer), Process_1kwzimz (FinTrust)

Worker registration verification

On startup, the Spring Boot log confirms all 51 workers subscribed to Zeebe. Key subscriptions verified:

```
Starting job worker: JobWorkerValue{type='processOrder', ...}
Starting job worker: JobWorkerValue{type='financeRequest', ...}
Starting job worker: JobWorkerValue{type='IMS', ...}
Starting job worker: JobWorkerValue{type='maintenaceLog', ...}
Starting job worker: JobWorkerValue{type='6Installment', ...}
Started WorkerApplication in 2.673 seconds (process running for 3.006)
```


3. Happy Path Test Cases

TC-01 — Post-Transaction IMS Deduction

Field	Detail
Test ID	TC-01
Path name	POSTransaction → IMS deduction
Trigger	Zeebe message POSTransaction
Start event	Event_1dkdh4c in Pro process
Message payload	orderId, productId="PROD-001", quantity=2, orderTotal=500.0, changeType="deduct", deductQty=2, currentStock=50
Workers expected	IMS → deductionIMS → IMS (display)
Expected outputs	imsUpdated=true, IMS=50, imsDeducted=true, newStockLevel=48
Expected result	Pro process instance COMPLETED, no incidents
Actual result	Pro instance 2251799814032912 COMPLETED in ~2 seconds
Worker log evidence	IMS: productId=PROD-001, changeType=deduct, quantity=2 / deductionIMS: productId=PROD-001, deductQty=2
Result	PASS

The screenshot displays the Camunda 8 web console interface. The top navigation bar includes 'Operate', 'Dashboard', 'Processes', 'Decisions', and 'Operations'. The main area shows the 'Instance History' for process instance '2251799814032912'. The 'Variables' tab is active, showing a list of variables and their values. A red box highlights the following variables:

Name	Value
IMS	50
changeType	"deduct"
currentStock	50
deductQty	2
imsDeducted	true
imsUpdateReference	"IMS-611CFC95"
imsUpdated	true
imsUpdatedAt	"2026-05-06T22:29:35.071095900Z"
newStockLevel	48

TC-02 — FixPro Tool Maintenance Cycle

Field	Detail
Test ID	TC-02

Field	Detail
Path name	sendFixPro → FixPro maintenance
Trigger	Zeebe message <code>sendFixPro</code>
Start event	<code>SE_FixPro</code> in FixPro process (<code>Process_0c6d9wt</code>)
Message payload	<code>toolId="TOOL-001", damageReport="Blade worn, requires sharpening", serviceReferenceNumber="FXP-DEMO-002", maintenanceType="repair", toolRepair="repair"</code>
Workers expected	<code>maintenanceLog</code> (logged on Pro path), then user tasks in FixPro subprocess: PAT test, service report, service label, handover form
End worker	<code>servedTools</code> (Message End Event in FixPro)
Expected result	FixPro process instance COMPLETED after user completes all user tasks in Tasklist
Actual result	FixPro instance <code>2251799814034868</code> COMPLETED after all Tasklist forms completed
User tasks completed	PATTest form, serviceReport form (Service Type: Repaired)
Result	PASS

Operation Type	Entity Type	Entity Key	Actor	Date
Complete	User task	2251799814045033	demo	2026-05-06 23:47:10
Assign	User task	2251799814045033	demo	2026-05-06 23:46:04
Complete	User task	2251799814044405	demo	2026-05-06 23:45:52
Assign	User task	2251799814044405	demo	2026-05-06 23:45:26
Complete	User task	2251799814034898	demo	2026-05-06 23:45:07
Assign	User task	2251799814034898	demo	2026-05-06 23:44:48

TC-03 — Serviced Tools Stock Update

Field	Detail
Test ID	TC-03
Path name	servedTools message → updateStock
Trigger	Zeebe message <code>servedTools</code>

Field	Detail
Start event	Event_1t98k2n in Pro process
Message payload	toolId="TOOL-001", serviceReferenceNumber="FXP-DEMO-001", productId="PROD-001", stockDelta=1, currentStock=50, replaceTools="no"
Workers expected	updateStock
Expected outputs	imsUpdated=true, newStockLevel=51, updateReference (UPD-XXXXXXXX)
Expected result	Pro process instance COMPLETED, no incidents
Actual result	Pro instance 2251799814032961 COMPLETED in ~2 seconds
Worker log evidence	updateStock: productId=PROD-001, stockDelta=1
Result	PASS

Pro

Process Instance Key

2251799814032961

Version

6

Start Date

2026-05-06 23:29:36

End Date

2026-05-06 23:29:36

Called Instances

None

Delete

Instance History

End date

Execution count

Pro

collect the serviced tools and service report

read the service report

does tool need replacing?

update stock

stock up the serviced tool

Gateway_1q9q4dm

Event_0tcplnw

Variables

Listeners

Operations Log

Name	Value
currentStock	50
imsUpdated	true
newStockLevel	51
productId	"PROD-001"
replaceTools	"no"
serviceReferenceNumber	"FXP-DEMO-001"
stockDelta	1
toolId	"TOOL-001"
updateReference	"UPD-220BA688"
updatedAt	"2026-05-06T22:29:36.060119100Z"

Add Variable

Copy variables

TC-04 — FinTrust Finance Installment Plan

Field	Detail
Test ID	TC-04
Path name	financeRequest → FinTrust 6-month plan
Trigger	Zeebe message financeRequest
Start event	Message start event in FinTrust process (Process_1kwzimz)
Message payload	orderTotal=1200.0, financeInstallments=6, customerId="CUST-001", financeRequestId="FIN-DEMO-001", customerEmail="test@probuild.com"

Field	Detail
Workers expected	6Installment → financeEmail (×6) → financeConfirmation → installmentCompletion
Expected outputs	monthlyAmount=210.0, totalWithInterest=1260.0, financeCompleted=true
Expected result	FinTrust process COMPLETED, all installment emails sent
Actual result	FinTrust instance 2251799814216310 COMPLETED — all 6 installment confirmations received, financeCompleted=true
Worker log evidence	6Installment: orderTotal=1200.0 / financeEmail: email=test@probuild.com, installment=1/6
Result	PASS

The screenshot shows the Camunda 8 Operate interface for process instance 2251799814216310. The process is 'Process_1kwzimz' and is in a 'COMPLETED' state. The 'Instance History' tab is selected, showing a list of events and activities. The 'Variables' tab is also visible, showing a list of variables and their values.

Variable	Value
completedAt	"2026-05-07T02:38:10.254544600Z"
customerEmail	"test@probuild.com"
customerId	"CUST-001"
emailSent	true
emailSentAt	"2026-05-07T02:29:21.998407400Z"
financeCompleted	true
financeConfirmation	true
financeConfirmedAt	"2026-05-07T02:26:38.128118500Z"
financeInstallments	6
financeRequestId	"FIN-DEMO-TC04"
installmentComplete	"FIN-DEMO-TC04"
monthlyAmount	210.0

TC-05 — Supplier Delivery Receipt & Quality Check

Field	Detail
Test ID	TC-05
Path name	Delivery message → barcode scan → quality inspect → auditID
Trigger	Zeebe message Delivery
Start event	Event_0xfbvlt in Pro process
Message payload	supplierId="SUP-001", orderId="ORD-SUPPLIER-001", productId="PROD-001", customerId="CUST-001", idDocumentType="passport"

Field	Detail
Workers expected	fetchDelivery → [barcode user task] → auditID
Expected outputs	deliveryStatus="IN_TRANSIT", goodsShipped=true, idVerified=true, auditPassed=true
Expected result	Pro process COMPLETED after barcode user task completed in Tasklist
Actual result	Pro instance 2251799814028750 COMPLETED after barcode form submitted and qualityInspect variable resolved
Worker log evidence	fetchDelivery: supplierId=SUP-001, orderId=ORD-SUPPLIER-001 / auditID: customerId=CUST-001
Result	PASS

✓ Pro Process Instance Key: 2251799814028750 Version: 6 Start Date: 2026-05-06 23:21:43 End Date: 2026-05-06 23:39:08 Called Instances: None Delete

Instance History		Variables	Listeners	Operations Log
☑	Pro			
☑	🔄 receive suppliers good	customerId		"CUST-001"
☑	⚙️ fetch goods and receipt	deliveryId		" "
☑	👤 scan barcode	deliveryStatus		"IN_TRANSIT"
☑	📋 perform quality inspection check	estimatedArrival		"2026-05-09"
☑	💎 qualityInspect	fastenersQuantity		null
☑	⚙️ update the IMS levels	heavyDutyDrillQuantity		null
☑	📋 stock up new good	idDocumentType		"passport"
☑	💎 Gateway_1v3pb7m	imsUpdateReference		"IMSU-3A4D3211"
☑	🕒 Event_0y99gJ5	imsUpdated		true
		imsUpdatedAt		"2026-05-06T22:39:08.76907900Z"
		insulationQuantity		null
		orderId		"ORD-SUPPLIER-001"
		plasterboardQuantity		null

Add Variable + Copy variables 📄

TC-06 — Full Customer Hire Flow (cross-pool)

Field	Detail
Test ID	TC-06
Path name	Customer website → Pro full hire flow
Trigger	POST /v2/process-instances — customer process (Process_1hz3vp5) started directly
Start event	Plain start event Event_09qssh7 in customer process
Initial variables	customerId, shoppingChoice="online", mainSelection="hire", isMember=true, productId, rentalDays=3, customerEmail, deliveryAddress

Field	Detail
Customer process flow	Start → manual tasks (auto-pass) → warehouse.IMS → tradeDatabase → Website form user task → sendOrder service end event
Website form	Completed via POST /v2/user-tasks/{key}/completion with hire variables
Cross-pool trigger	sendOrder worker publishes Zeebe message sendOrder → Pro process Event_0sh42cc creates new instance
Pro workers fired	processOrder → customerIMSAvail → POSTtransaction
Expected result	Customer process COMPLETED, Pro process instance COMPLETED
Actual result	Customer instance 2251799814249622 COMPLETED; Pro instance 2251799814250380 COMPLETED
Result	PASS

Operate

DashboardProcessesDecisionsOperations

Non-production licenseNon-commercial license

Pro

Process Instance Key2251799814250380

Version9

Start Date2026-05-08 03:30:00

End Date2026-05-08 03:30:00

Called InstancesNone

Delete

Instance History

End dateExecution count

recieve order

process order

Gateway_1kxguq1

Is customer a membership owner?

give membership benefits

Gateway_1mv0on1

Is tool available?

availability?

recieve customer address

did customer pay finance or

Process Payment (Online)

Gateway_15342p4

get rental duration

how long is the rental?

ship tool to customer

Event_1rqy2zw

start post transaction process

send email requesting delivery of rental goods

Variables

Listeners

Operations Log

orderCapturedtrue

orderCapturedAt"2026-05-07T02:59:32.003781100Z"

orderChannel"online"

orderId"ORD-FF4E217A"

orderStatus"RECEIVED"

orderTotal300.0

paymentProcessedAt"2026-05-07T03:02:01.270402700Z"

paymentSuccessful"yes"

posProcessedAt"2026-05-07T03:03:35.452428500Z"

posTransactionId"POS-20159348"

postcode"BS1 1AA"

productId"PROD-001"

Add VariableCopy variables

4. Defect Log

All defects below were discovered during test execution and resolved before the test pass was recorded.

DEF-01 — IMS gateway variable name mismatch

Field	Detail
Defect ID	DEF-01
Severity	High — caused process incident, blocked TC-01
Element	Gateway_09chtjp in Pro process
Error type	EXTRACT_VALUE_ERROR

Field	Detail
Error message	Expected result of the expression 'IMS >= 10' to be 'BOOLEAN', but was 'NULL'. No variable found with name 'IMS'
Root cause	The IMS service task worker (<code>imsUpdate()</code> in <code>InventoryWorkers.java</code>) output the stock level as <code>currentStock</code> but the downstream BPMN gateway condition used <code>=IMS >= 10</code> , expecting a variable literally named IMS . The variable IMS was never set in the process scope.
Fix applied	Added <code>result.put("IMS", 50)</code> to the <code>imsUpdate()</code> method in <code>InventoryWorkers.java</code> . The worker now outputs both <code>currentStock</code> and IMS with the same value.
Verification	TC-01 re-run after fix — Pro instance COMPLETED with no incidents.
Status	RESOLVED

DEF-02 — replaceTools gateway condition failure

Field	Detail
Defect ID	DEF-02
Severity	Medium — caused process incident, blocked TC-03
Element	<code>replaceTools</code> (inclusiveGateway) in Pro process
Error type	<code>CONDITION_ERROR</code>
Error message	Expected at least one condition to evaluate to true, or to have a default flow
Root cause	The gateway conditions check <code>=replaceTools = "yes"</code> and <code>=replaceTools = "no"</code> . The preceding task (<code>Activity_06gwxr8</code> , "read the service report") is a <code>bpmn:task</code> (abstract/pass-through in Zeebe), which executes with no user interaction and sets no variables. The variable <code>replaceTools</code> is therefore always null when the gateway evaluates.
Fix applied	For demo purposes: <code>replaceTools="no"</code> included in the message publication payload so the variable is in scope when the gateway evaluates. Permanent fix (pending BPMN update): convert the preceding task to a <code>bpmn:userTask</code> with a form field that sets <code>replaceTools</code> .
Verification	TC-03 re-run with <code>replaceTools="no"</code> in payload — Pro instance COMPLETED with no incidents.
Status	RESOLVED (workaround)

DEF-03 — qualityInspect gateway condition failure

Field	Detail
Defect ID	DEF-03

Field	Detail
Severity	Medium — caused process incident mid-TC-05
Element	<code>qualityInspect</code> (inclusiveGateway) in Pro process
Error type	<code>CONDITION_ERROR</code>
Error message	Expected at least one condition to evaluate to true, or to have a default flow
Root cause	Same pattern as DEF-02. The preceding task (<code>qualityInspection</code> , "perform quality inspection check") is a <code>bpmn:manualTask</code> . Manual tasks in Camunda 8 (Zeebe) pass through automatically with no variable output. The variable <code>qualityInspect</code> is never set.
Fix applied	Variable <code>qualityInspect="pass"</code> set on the stalled element instance via <code>PUT /v2/element-instances/{key}/variables</code> , then the incident resolved via <code>POST /v2/incidents/{key}/resolution</code> . Permanent fix: convert <code>qualityInspection</code> to a <code>bpmn:userTask</code> with a pass/fail form field.
Verification	TC-05 instance continued and reached COMPLETED after incident resolved.
Status	RESOLVED (API workaround)

DEF-04 — toolRepair gateway condition failure (FixPro)

Field	Detail
Defect ID	DEF-04
Severity	Medium — caused process incident, blocked TC-02 first run
Element	<code>toolRepair</code> (exclusiveGateway) in FixPro subprocess (<code>Process_0c6d9wt</code>)
Error type	<code>CONDITION_ERROR</code>
Error message	Expected at least one condition to evaluate to true, or to have a default flow
Root cause	Two <code>bpmn:manualTask</code> elements precede this gateway ("Read the digital handover review" and "inspect damages from the tool"). Both pass through without setting any variables. The gateway checks <code>=toolRepair = "repair"</code> , <code>=toolRepair = "maintenace"</code> (note: BPMN typo — "maintenace" not "maintenance"), and <code>=toolRepair = "dispose"</code> .
Fix applied	<code>toolRepair="repair"</code> included in message payload. Permanent fix: convert one of the manual tasks to a user task with a select field for repair type.
Verification	TC-02 re-run with <code>toolRepair="repair"</code> — FixPro instance COMPLETED after user tasks in Tasklist.
Status	RESOLVED (workaround)

DEF-05 — Incorrect REST API field name for message publication

Field	Detail
Defect ID	DEF-05
Severity	Low — blocked all message-triggered tests until resolved
Error	400 Bad Request: "Request property [messageName] cannot be parsed"
Root cause	The Camunda 8 REST API v2 message publication endpoint (<code>POST /v2/messages/publication</code>) uses the field name <code>name</code> for the message name — not <code>messageName</code> as in the v1 API.
Fix applied	Changed all API calls from <code>"messageName": "..."</code> to <code>"name": "..."</code> .
Verification	All five TC-01 through TC-05 messages published successfully (<code>200 OK</code> with <code>messageKey</code>).
Status	RESOLVED

DEF-06 — FinTrust installmentComplete correlation key null

Field	Detail
Defect ID	DEF-06
Severity	Medium — blocked TC-04, caused incident on message catch event
Element	<code>Event_0dka0q8</code> ("confirm installmt paid") in FinTrust process
Error type	<code>EXTRACT_VALUE_ERROR</code>
Error message	Failed to extract the correlation key for 'installmentComplete': The value must be either a string or a number, but was 'NULL'
Root cause	The message intermediate catch event correlates on <code>=installmentComplete</code> , but no worker or task in the upstream flow sets this variable. The <code>financeRequestId</code> is available but <code>installmentComplete</code> is never initialised.
Fix applied	Variable <code>installmentComplete</code> set to the value of <code>financeRequestId</code> ("FIN-DEMO-TC04") via <code>PUT /v2/element-instances/{key}/variables</code> , incident resolved, then <code>installmentComplete</code> message published with matching correlation key.
Verification	TC-04 instance proceeded past the catch event and continued through all 6 installments to COMPLETED.
Status	RESOLVED (workaround) — permanent fix: add an upstream service task or script task to set <code>installmentComplete = financeRequestId</code> before the catch event.

5. Test Summary

ID	Test case	Result
TC-01	POSTransaction → IMS deduction	PASS
TC-02	sendFixPro → FixPro maintenance cycle	PASS

ID	Test case	Result
TC-03	servicedTools → updateStock	PASS
TC-04	financeRequest → FinTrust installments	PASS
TC-05	Delivery → barcode → auditID	PASS
TC-06	Customer Website → Pro hire flow	PASS

ID	Defect	Status
DEF-01	IMS gateway variable name mismatch	RESOLVED
DEF-02	replaceTools gateway — manual task sets no variable	RESOLVED (workaround)
DEF-03	qualityInspect gateway — manual task sets no variable	RESOLVED (workaround)
DEF-04	toolRepair gateway — manual tasks set no variable	RESOLVED (workaround)
DEF-05	REST API field name <code>messageName</code> vs <code>name</code>	RESOLVED
DEF-06	FinTrust <code>installmentComplete</code> correlation key null	RESOLVED (workaround)
DEF-07	<code>customerIMSAvail</code> output name mismatch — gateway expected <code>onlineavailability</code> , worker output <code>availableQty</code>	RESOLVED
DEF-08	Hire path gateways <code>hirePayment</code> and <code>financeDuration</code> null — variables were omitted from test API call (form already has correct field keys)	RESOLVED

6/6 test cases pass. 8/8 defects resolved.

DEF-02, DEF-03, DEF-04, DEF-07, and DEF-08 all arise from variables expected by gateways that are either not set by the preceding task, or are named differently than what the worker outputs. DEF-06 reflects a missing initialisation upstream of the FinTrust message catch event.

6. Regression Gate

- No new BPMN `zeebe:taskDefinition` type may be added without a corresponding `@JobWorker` and an entry in this document.
- Any change to a process variable name must update `service-task-traceability.md` and all workers that read or write it.
- Before each demo: verify all processes listed in Section 1 are deployed and executable in Operate.
- Cross-reference: `docs/service-task-traceability.md` tracks all 51 worker types against their BPMN element.